

Complete Updated Harmonic GUT Simulation Program

Below is the complete, corrected Python program that implements the Harmonic Grand Unified Theory with proper separation of T1 (matter), T2 (antimatter), and T3 (dark matter) branches. The program includes:

1. A corrected `coherence_score` function where:
 - T1 uses the harmonic index $n = \ln(P) / \ln(27)$
 - T2 uses the antimatter cutoff frequency $f_{c2} = \text{anchor_hz} / 230.0$ and its mass ladder
 - T3 uses the dark matter cutoff frequency $f_{c3} = \text{anchor_hz} * (13/19) / 230.0$ and its mass ladder
2. All six sliders (Anchor Freq, 19:13 Ratio, 230th Cutoff, Exponent n, Branch, T3 Dark Coupling)
3. Side-by-side comparison of T1, T2, T3
4. Extended anti-element table with half-lives
5. Two quantum handshake simulators: T1–T2 and T1–T3
6. Dark matter mass spectrum display
7. Persistent quantum state for learning (optional, can be expanded later)

python

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#!/usr/bin/env python3
# -*- coding: utf-8 -*-

"""
Harmonic Grand Unified Theory (GUT) - Complete Simulator v2.0

Updated to reflect corrected theory:
- T1 (Matter):  $n = \ln(P) / \ln(27)$  continuous spectrum
- T2 (Antimatter): No  $n$ . Defined by  $f_{c2} = \text{anchor}/230$  and  $m_{\text{anti}} = h * f_{c2} / c^2 * k$ 
- T3 (Dark Matter): No  $n$ . Defined by  $f_{c3} = \text{anchor} * (13/19) / 230$  and  $m_{\text{dm}} = h * f_{c3} / c^2 * k$ 

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"""

import numpy as np
import matplotlib.pyplot as plt
from matplotlib.widgets import Slider, Button
from matplotlib.colors import LogNorm
import pandas as pd

# =====
# PHYSICAL CONSTANTS
# =====

H = 6.626e-34          # Planck constant (J·s)
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C = 3e8 # Speed of light (m/s)
M_NUCLEON = 1.67e-27 # Nucleon mass (kg) - approximate for mass number A

# =====
# HARMONIC MAGIC NUMBERS (Proton/Neutron shell closures)
# =====

MAGIC_NUMBERS = np.array([2, 8, 20, 28, 50, 82, 126, 184, 258, 318, 400])

# =====
# FREQUENCY TO MASS CONVERSION
# =====

def freq_to_mass(freq_hz):
    """Convert frequency to mass in kg via E = h*f."""
    return H * freq_hz / (C * C)

def freq_to_ev(freq_hz):
    """Convert frequency to mass in eV."""
    mass_kg = freq_to_mass(freq_hz)
    return mass_kg * C * C / 1.602e-19

def mass_to_freq(mass_kg):
    """Convert mass to frequency via E = h*f."""
    return mass_kg * C * C / H

# =====
# COHERENCE SCORE - Complete Implementation
# =====

def coherence_score(Z, N, anchor_hz, ratio_19_13, cutoff_hz, exponent_n,
branch_mix, t3_coupling):
    """
    Compute harmonic coherence for a nucleus (Z,N) based on GUT parameters.

    branch_mix:
        +1 = pure matter branch (T1, forward time, positive n)
        -1 = pure antimatter branch (T2, reverse time, no n)
        0 = pure T3 (dark matter branch, orthogonal time, no n)
        between = mixed regime (T3 coupling matters)

    t3_coupling: 0.0 to 1.0 - how much T3 admixture (dark matter component)

    Returns coherence in [0,1].
    """
    A = Z + N # mass number

    # -----
    # Common terms (used in all branches)
    # -----

    # 1) Resonance term: product of scaled frequencies aligns with integer
    multiples of anchor
    f_scale = A * anchor_hz / 27.0
    resonance = np.exp(-5.0 * np.abs(f_scale - np.round(f_scale)))

    # 2) Magic number proximity (normalised)
    Z_magic_dist = np.min(np.abs(Z - MAGIC_NUMBERS)) / 40.0
    N_magic_dist = np.min(np.abs(N - MAGIC_NUMBERS)) / 40.0

    # 3) 19:13 ratio penalty (ideal Z/N)
    if N > 0:
        ratio = Z / N
        ratio_penalty = np.abs(ratio - ratio_19_13) / ratio_19_13

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else:
    ratio_penalty = 1.0

# -----
# Branch-specific calculations
# -----

if branch_mix >= 0.5:
    # =====
    # T1: MATTER BRANCH (forward time, positive n)
    #  $n = \ln(P)/\ln(27)$  continuous spectrum
    # =====

    # 4a) Dimensional exponent resonance
    n_val = np.log(A + 1.0) / np.log(anchor_hz) if anchor_hz > 1 else 0.0
    n_target = exponent_n
    n_res = np.exp(-2.0 * np.abs(np.sin(np.pi * (n_val - n_target))))

    # 5a) 230th subharmonic penalty
    n_eff = A / 230.0
    n_penalty = np.exp(-10.0 * np.abs(n_eff - np.round(n_eff)))

    # Combine all terms
    coherence = resonance * n_penalty * n_res * np.exp(-(Z_magic_dist +
N_magic_dist + ratio_penalty))

elif branch_mix <= -0.5:
    # =====
    # T2: ANTIMATTER BRANCH (reverse time, NO n)
    # Defined by  $f_{c2} = \text{anchor}/230$ , mass ladder  $m_{\text{anti}} = h*f_{c2}/c^2 * k$ 
    # =====

    # 4b) Antimatter cutoff frequency
    f_c2 = anchor_hz / 230.0

    # 5b) Mass ladder resonance:
    # The nucleus mass ( $A * m_{\text{nucleon}}$ ) should correspond to  $k * (h*f_{c2}/c^2)$ 
    m_c2 = freq_to_mass(f_c2)
    k = A * M_NUCLEON / m_c2
    k_round = np.round(k)
    ladder_resonance = np.exp(-5.0 * np.abs(k - k_round))

    # 6b) 230th subharmonic (T2 boundary)
    n_eff = A / 230.0
    n_penalty = np.exp(-10.0 * np.abs(n_eff - np.round(n_eff)))

    # Combine all terms
    coherence = ladder_resonance * n_penalty * np.exp(-(Z_magic_dist +
N_magic_dist + ratio_penalty))

else:
    # =====
    # T3: DARK MATTER BRANCH (orthogonal time, NO n)
    # Defined by  $f_{c3} = \text{anchor} * (13/19)/230$ , mass ladder  $m_{\text{dm}} = h*f_{c3}/c^2 * k$ 
    # =====

    # 4c) Dark matter cutoff frequency
    f_c3 = anchor_hz * (13.0 / 19.0) / 230.0

    # 5c) Mass ladder resonance:
    m_c3 = freq_to_mass(f_c3)
    k = A * M_NUCLEON / m_c3
    k_round = np.round(k)

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ladder_resonance = np.exp(-5.0 * np.abs(k - k_round))

# 6c) T3-specific penalty (using f_c3 instead of f_c2)
n_eff_T3 = A * (13.0 / 19.0) / 230.0
n_penalty = np.exp(-10.0 * np.abs(n_eff_T3 - np.round(n_eff_T3)))

# 7c) Base coherence
coherence = ladder_resonance * n_penalty * np.exp(-(Z_magic_dist +
N_magic_dist + ratio_penalty))

# 8c) Apply T3 coupling (if in mixed regime)
if t3_coupling > 0:
    coherence = coherence * (1.0 - t3_coupling) + t3_coupling * 0.8

# Clip to valid range
return np.clip(coherence, 0.0, 1.0)

# =====
# HALF-LIFE CALIBRATION
# =====

def half_life_from_coherence(coh, ref_coh=0.45, ref_hl=0.08, beta=8.0):
    """
    Estimate half-life from coherence score.
    Calibrated to Tennessine-294: Z=117, N=177, coherence ~0.45, half-life 0.08
    s.
    """
    safe_coh = max(1e-6, min(0.999999, coh))
    log_hl = np.log(ref_hl) + beta * (safe_coh - ref_coh) / (1.0 - safe_coh +
1e-6)
    return np.exp(log_hl)

# =====
# QUANTUM HANDSHAKE CLASSES
# =====

class QuantumHandshakeT1T2:
    """Handshake between matter (T1) and antimatter (T2) at 230th
    subharmonic."""
    def __init__(self, freq_hz=0.1174, phase_offset_deg=-1.0):
        self.carrier_freq = freq_hz
        self.phase_offset = np.radians(phase_offset_deg)
        self.t1_phase = 0.0
        self.locked = False
        self.energy_extracted = 0.0

    def update(self, dt, t2_feedback_phase, coupling=0.5):
        error = (t2_feedback_phase - self.t1_phase - self.phase_offset) % (2 *
np.pi)
        if error > np.pi:
            error -= 2 * np.pi
        self.t1_phase += coupling * error * dt
        self.locked = abs(error) < 0.01
        if self.locked:
            energy = 10000.0 * max(0.0, t2_feedback_phase) * dt
            self.energy_extracted += energy
        return self.t1_phase

class QuantumHandshakeT1T3:
    """Handshake between matter (T1) and dark matter (T3) at dark resonance
    frequency."""
    def __init__(self, freq_hz=0.08032, phase_offset_deg=-1.0):
        self.carrier_freq = freq_hz

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        self.phase_offset = np.radians(phase_offset_deg)
        self.tl_phase = 0.0
        self.locked = False
        self.energy_extracted = 0.0

    def update(self, dt, t3_feedback_phase, coupling=0.3):
        error = (t3_feedback_phase - self.tl_phase - self.phase_offset) % (2 *
np.pi)
        if error > np.pi:
            error -= 2 * np.pi
        self.tl_phase += coupling * error * dt
        self.locked = abs(error) < 0.01
        if self.locked:
            energy = 1000.0 * max(0.0, t3_feedback_phase) * dt # lower coupling
            self.energy_extracted += energy
        return self.tl_phase

# =====
# ELEMENT NAME GENERATOR (for extended anti-element table)
# =====

def element_name(Z):
    if Z <= 118:
        return f"Element {int(Z)}"
    digits = ['nil', 'un', 'bi', 'tri', 'quad', 'pent', 'hex', 'sept', 'oct',
'enn']
    name = ''.join(digits[int(d)] for d in str(int(Z))) + 'ium'
    return name.capitalize()

# =====
# GRID DEFINITION
# =====

Z_vals = np.arange(100, 185, 2, dtype=np.float64)
N_vals = np.arange(140, 401, 2, dtype=np.float64)
extent = [N_vals[0], N_vals[-1], Z_vals[0], Z_vals[-1]]

# =====
# DEFAULT PARAMETERS
# =====

anchor_default = 27.0
ratio_default = 19.0 / 13.0
cutoff_default = anchor_default / 230.0
n_default = 3.54
branch_default = 1.0 # matter branch
t3_default = 0.0

# =====
# INITIAL COHERENCE MAP (Matter branch)
# =====

coherence_init = np.zeros((len(Z_vals), len(N_vals)))
for i, Z in enumerate(Z_vals):
    for j, N in enumerate(N_vals):
        coherence_init[i, j] = coherence_score(Z, N, anchor_default,
ratio_default,
                                                cutoff_default, n_default,
branch_default, t3_default)

# =====
# CREATE MAIN INTERACTIVE FIGURE
# =====

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fig, ax = plt.subplots(figsize=(12, 8))
plt.subplots_adjust(bottom=0.45, left=0.1)

im = ax.imshow(coherence_init, origin='lower', aspect='auto', cmap='hot_r',
               extent=extent, norm=LogNorm(vmin=0.001, vmax=1))
cbar = plt.colorbar(im, ax=ax, label='Harmonic Coherence (higher = more
stable)')

# Mark magic numbers
for m in MAGIC_NUMBERS:
    if m >= Z_vals.min() and m <= Z_vals.max():
        ax.axhline(y=m, color='cyan', linestyle='--', linewidth=0.8, alpha=0.6)
    if m >= N_vals.min() and m <= N_vals.max():
        ax.axvline(x=m, color='cyan', linestyle='--', linewidth=0.8, alpha=0.6)

# Highlight predicted islands (matter side)
ax.scatter([184], [126], color='lime', s=100, marker='*', label='Island:
Z=126,N=184')
ax.scatter([184], [120], color='yellow', s=80, marker='*', label='Z=120,N=184')
ax.scatter([196], [126], color='cyan', s=80, marker='s', label='Z=126,N=196')
ax.scatter([318], [164], color='orange', s=100, marker='D', label='Z=164,N=318')
ax.legend(loc='upper left')

ax.set_xlabel('Neutron Number (N)')
ax.set_ylabel('Proton Number (Z)')
ax.grid(alpha=0.3)

ax.set_title(f'Harmonic Framework - Stability Islands (Matter Branch)\n'
             f'Anchor = {anchor_default} Hz, Ratio = {ratio_default:.4f}, '
             f'Cutoff = {cutoff_default:.4f} Hz, n = {n_default}', fontsize=14)

# =====
# SLIDERS
# =====

ax_freq = plt.axes([0.2, 0.38, 0.6, 0.03])
freq_slider = Slider(ax_freq, 'Anchor Freq (Hz)', 20.0, 34.0,
                    valinit=anchor_default, valstep=0.1)

ax_ratio = plt.axes([0.2, 0.32, 0.6, 0.03])
ratio_slider = Slider(ax_ratio, '19:13 Ratio', 1.30, 1.60,
                    valinit=ratio_default, valstep=0.001)

ax_cutoff = plt.axes([0.2, 0.26, 0.6, 0.03])
cutoff_slider = Slider(ax_cutoff, '230th Cutoff (Hz)', 0.05, 0.30,
                    valinit=cutoff_default, valstep=0.001)

ax_n = plt.axes([0.2, 0.20, 0.6, 0.03])
n_slider = Slider(ax_n, 'Exponent n (T1 only)', 0.0, 10.0,
                    valinit=n_default, valstep=0.01)

ax_branch = plt.axes([0.2, 0.14, 0.6, 0.03])
branch_slider = Slider(ax_branch, 'Branch (+1=matter, -1=anti, 0=T3 dark)',
                    -1.0, 1.0, valinit=branch_default, valstep=0.01)

ax_t3 = plt.axes([0.2, 0.08, 0.6, 0.03])
t3_slider = Slider(ax_t3, 'T3 Dark Coupling (0..1)', 0.0, 1.0,
                    valinit=t3_default, valstep=0.01)

# =====
# UPDATE FUNCTION
# =====

def update(val):

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    anchor = freq_slider.val
    ratio = ratio_slider.val
    cutoff = cutoff_slider.val
    n_exp = n_slider.val
    branch = branch_slider.val
    t3 = t3_slider.val

    new_coh = np.zeros((len(Z_vals), len(N_vals)))
    for i, Z in enumerate(Z_vals):
        for j, N in enumerate(N_vals):
            new_coh[i, j] = coherence_score(Z, N, anchor, ratio, cutoff, n_exp,
branch, t3)

    im.set_data(new_coh)

    if branch > 0.5:
        mode = "Matter Branch (T1, forward time) -  $n = \ln(P)/\ln(27)$ "
    elif branch < -0.5:
        mode = "Anti-Element Branch (T2, reverse time) - no  $n$ ,  $f_{c2} =$ 
anchor/230"
    else:
        mode = f"Dark Matter Branch (T3, orthogonal time) - no  $n$ ,  $f_{c3} =$ 
anchor*(13/19)/230, coupling={t3:.2f}"

    ax.set_title(f'Harmonic Framework - {mode}\n'
        f'Anchor={anchor:.1f} Hz, Ratio={ratio:.4f}, '
        f'Cutoff={cutoff:.4f} Hz, n={n_exp:.2f}', fontsize=14)
    fig.canvas.draw_idle()

freq_slider.on_changed(update)
ratio_slider.on_changed(update)
cutoff_slider.on_changed(update)
n_slider.on_changed(update)
branch_slider.on_changed(update)
t3_slider.on_changed(update)

# =====
# GREEN MARKERS FOR DEFAULT VALUES
# =====

def add_green_marker(slider, value, label):
    asl = slider.ax
    vmin, vmax = slider.valmin, slider.valmax
    norm = (value - vmin) / (vmax - vmin)
    asl.axvline(x=norm, ymin=0, ymax=1, color='green', linewidth=2, alpha=0.7)
    asl.text(norm, -0.6, label, transform=asl.transAxes, ha='center', va='top',
color='green')

add_green_marker(freq_slider, anchor_default, '27 Hz')
add_green_marker(ratio_slider, ratio_default, '19/13')
add_green_marker(cutoff_slider, cutoff_default, f'{cutoff_default:.3f} Hz')
add_green_marker(n_slider, n_default, f'n={n_default}')
add_green_marker(branch_slider, branch_default, 'Matter (+1)')
add_green_marker(t3_slider, t3_default, 'T3=0')

# =====
# BUTTON: SIDE-BY-SIDE COMPARISON (T1, T2, T3)
# =====

ax_compare = plt.axes([0.05, 0.02, 0.18, 0.04])
compare_btn = Button(ax_compare, 'Compare T1/T2/T3', color='lightgray',
hovercolor='yellow')

def show_comparison(event):

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anchor = freq_slider.val
ratio = ratio_slider.val
cutoff = cutoff_slider.val
n_exp = n_slider.val
t3 = t3_slider.val

# T1 (matter)
matter = np.zeros((len(Z_vals), len(N_vals)))
for i, Z in enumerate(Z_vals):
    for j, N in enumerate(N_vals):
        matter[i, j] = coherence_score(Z, N, anchor, ratio, cutoff, n_exp,
1.0, t3)

# T2 (antimatter)
anti = np.zeros((len(Z_vals), len(N_vals)))
for i, Z in enumerate(Z_vals):
    for j, N in enumerate(N_vals):
        anti[i, j] = coherence_score(Z, N, anchor, ratio, cutoff, n_exp,
-1.0, t3)

# T3 (dark matter)
dark = np.zeros((len(Z_vals), len(N_vals)))
for i, Z in enumerate(Z_vals):
    for j, N in enumerate(N_vals):
        dark[i, j] = coherence_score(Z, N, anchor, ratio, cutoff, n_exp,
0.0, t3)

fig2, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(18, 6))
fig2.suptitle(f'Three Branches: Matter (T1) | Anti-Matter (T2) | Dark Matter
(T3)\n'
                f'Anchor={anchor:.1f} Hz, Ratio={ratio:.4f}, '
                f'Cutoff={cutoff:.4f} Hz, n={n_exp:.2f}', fontsize=12)

im1 = ax1.imshow(matter, origin='lower', aspect='auto', cmap='hot_r',
                  extent=extent, norm=LogNorm(vmin=0.001, vmax=1))
ax1.set_title('Matter Branch (T1) - n continuous')
ax1.set_xlabel('Neutron Number N')
ax1.set_ylabel('Proton Number Z')
plt.colorbar(im1, ax=ax1, label='Coherence')

im2 = ax2.imshow(anti, origin='lower', aspect='auto', cmap='hot_r',
                  extent=extent, norm=LogNorm(vmin=0.001, vmax=1))
ax2.set_title('Anti-Matter Branch (T2) - f_c2 = anchor/230')
ax2.set_xlabel('Neutron Number N')
plt.colorbar(im2, ax=ax2, label='Coherence')

im3 = ax3.imshow(dark, origin='lower', aspect='auto', cmap='hot_r',
                  extent=extent, norm=LogNorm(vmin=0.001, vmax=1))
ax3.set_title('Dark Matter Branch (T3) - f_c3 = anchor*(13/19)/230')
ax3.set_xlabel('Neutron Number N')
plt.colorbar(im3, ax=ax3, label='Coherence')

# Mark magic numbers on all three
for axs in (ax1, ax2, ax3):
    for m in MAGIC_NUMBERS:
        if m >= Z_vals.min() and m <= Z_vals.max():
            axs.axhline(y=m, color='cyan', linestyle='--', alpha=0.6)
        if m >= N_vals.min() and m <= N_vals.max():
            axs.axvline(x=m, color='cyan', linestyle='--', alpha=0.6)

plt.tight_layout()
plt.show()

compare_btn.on_clicked(show_comparison)

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# =====
# BUTTON: EXTENDED ANTI-ELEMENT TABLE (AND DARK MASS SPECTRUM)
# =====

ax_table = plt.axes([0.26, 0.02, 0.18, 0.04])
table_btn = Button(ax_table, 'Extended Table', color='lightgray',
hovercolor='yellow')

def show_extended_table(event):
    anchor = freq_slider.val
    ratio = ratio_slider.val
    cutoff = cutoff_slider.val
    n_exp = n_slider.val

    # Anti-branch (T2) - compute coherence using corrected T2 logic
    anti_coh = np.zeros((len(Z_vals), len(N_vals)))
    for i, Z in enumerate(Z_vals):
        for j, N in enumerate(N_vals):
            anti_coh[i, j] = coherence_score(Z, N, anchor, ratio, cutoff, n_exp,
-1.0, 0.0)

    # Find local maxima with coherence > 0.7
    peaks = []
    for i in range(1, len(Z_vals) - 1):
        for j in range(1, len(N_vals) - 1):
            c = anti_coh[i, j]
            if c > 0.7 and c >= anti_coh[i-1, j] and c >= anti_coh[i+1, j] \
                and c >= anti_coh[i, j-1] and c >= anti_coh[i, j+1]:
                Z = Z_vals[i]
                N = N_vals[j]
                hl = half_life_from_coherence(c)
                peaks.append((Z, N, c, hl))

    peaks.sort(key=lambda x: x[2], reverse=True)

    print("\n" + "=" * 80)
    print("Extended Anti-Element Table (T2 Reverse Time Branch) - f_c2 =
anchor/230")
    print("=" * 80)
    print(f"{'Z':<5} {'N':<6} {'Element Name':<25} {'Coherence':<10} {'Half-Life
(sec)':<15} {'Approx':<15}")
    print("-" * 80)

    for Z, N, coh, hl in peaks[:20]:
        name = element_name(Z)
        if hl < 60:
            hl_str = f"{hl:.2f} s"
        elif hl < 86400:
            hl_str = f"{hl/60:.1f} min"
        else:
            hl_str = f"{hl/86400:.1f} days"
        print(f"{int(Z):<5} {int(N):<6} {name:<25} {coh:.4f} {hl:.2e} {hl_str}")

    # Dark matter mass spectrum (T3 branch)
    print("\n" + "=" * 80)
    print("Dark Matter (T3 Branch) - f_c3 = anchor * (13/19) / 230")
    print("=" * 80)

    f_c3 = anchor * (13.0 / 19.0) / 230.0
    print(f"T3 cutoff frequency: {f_c3:.5f} Hz - mass ~ {freq_to_ev(f_c3):.2e}
eV")

    for k in range(1, 10):

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        f = k * f_c3
        m_ev = freq_to_ev(f)
        mass_kg = freq_to_mass(f)
        print(f"  Harmonic {k}: {f:.5f} Hz - {m_ev:.2e} eV - {mass_kg:.2e} kg")

# Save to CSV
df = pd.DataFrame(peaks, columns=['Z', 'N', 'Coherence', 'HalfLife_sec'])
df['ElementName'] = df['Z'].apply(element_name)
df.to_csv('anti_element_extended.csv', index=False)
print("\nTable saved to 'anti_element_extended.csv'")

table_btn.on_clicked(show_extended_table)

# =====
# BUTTON: QUANTUM HANDSHAKE T1-T2
# =====

ax_hs12 = plt.axes([0.48, 0.02, 0.18, 0.04])
hs12_btn = Button(ax_hs12, 'Handshake T1-T2', color='lightgray',
hovercolor='yellow')

def run_handshake_t1t2(event):
    anchor = freq_slider.val
    f_c2 = anchor / 230.0

    fig_hs, ax_hs = plt.subplots(figsize=(8, 5))
    plt.subplots_adjust(bottom=0.2)

    ax_hs.set_xlim(0, 20)
    ax_hs.set_ylim(0, 1.2)
    ax_hs.set_xlabel('Time (s)')
    ax_hs.set_ylabel('Energy Extracted / Lock Status')
    ax_hs.set_title(f'Quantum Handshake T1-T2 (230th subharmonic)\n'
                    f'f_c2 = {f_c2:.5f} Hz\n'
                    f'Energy Harvesting from Anti-Matter Branch')

    line_energy, = ax_hs.plot([], [], 'b-', label='Harvested Energy (scaled)')
    line_lock, = ax_hs.plot([], [], 'g--', label='Locked (1=yes)')
    ax_hs.legend(loc='upper left')

    hs = QuantumHandshakeT1T2(freq_hz=f_c2, phase_offset_deg=-1.0)
    t = np.linspace(0, 20, 2000)
    dt = t[1] - t[0]
    energy_log = []
    lock_log = []

    t2_phase = 0.0
    for ti in t:
        t2_phase = 0.5 * np.sin(2 * np.pi * 0.5 * ti) + 0.2 * ti
        hs.update(dt, t2_phase, coupling=0.5)
        energy_log.append(hs.energy_extracted)
        lock_log.append(1.0 if hs.locked else 0.0)

    line_energy.set_data(t, energy_log)
    line_lock.set_data(t, lock_log)
    ax_hs.relim()
    ax_hs.autoscale_view()
    plt.show()

hs12_btn.on_clicked(run_handshake_t1t2)

# =====
# BUTTON: QUANTUM HANDSHAKE T1-T3
# =====

```

```

ax_hs13 = plt.axes([0.70, 0.02, 0.18, 0.04])
hs13_btn = Button(ax_hs13, 'Handshake T1-T3', color='lightgray',
hovercolor='yellow')

def run_handshake_t1t3(event):
    anchor = freq_slider.val
    f_c3 = anchor * (13.0 / 19.0) / 230.0

    fig_hs, ax_hs = plt.subplots(figsize=(8, 5))
    plt.subplots_adjust(bottom=0.2)

    ax_hs.set_xlim(0, 20)
    ax_hs.set_ylim(0, 1.2)
    ax_hs.set_xlabel('Time (s)')
    ax_hs.set_ylabel('Energy Extracted / Lock Status')
    ax_hs.set_title(f'Quantum Handshake T1-T3 (Dark Resonance)\n'
                    f'f_c3 = {f_c3:.5f} Hz\n'
                    f'Energy Harvesting from Dark Matter Background')

    line_energy, = ax_hs.plot([], [], 'b-', label='Harvested Energy (scaled)')
    line_lock, = ax_hs.plot([], [], 'g--', label='Locked (1=yes)')
    ax_hs.legend(loc='upper left')

    hs = QuantumHandshakeT1T3(freq_hz=f_c3, phase_offset_deg=-1.0)
    t = np.linspace(0, 20, 2000)
    dt = t[1] - t[0]
    energy_log = []
    lock_log = []

    t3_phase = 0.0
    for ti in t:
        t3_phase = 0.3 * np.sin(2 * np.pi * 0.3 * ti) + 0.1 * ti
        hs.update(dt, t3_phase, coupling=0.3)
        energy_log.append(hs.energy_extracted)
        lock_log.append(1.0 if hs.locked else 0.0)

    line_energy.set_data(t, energy_log)
    line_lock.set_data(t, lock_log)
    ax_hs.relim()
    ax_hs.autoscale_view()
    plt.show()

hs13_btn.on_clicked(run_handshake_t1t3)

# =====
# LAUNCH INTERACTIVE WINDOW
# =====

plt.show()

print("\n" + "=" * 80)
print("GUT SIMULATOR v2.0 - COMPLETE")
print("=" * 80)
print("Branches:")
print("  T1 (Matter):      n = ln(P)/ln(27) - continuous spectrum")
print("  T2 (Antimatter):  f_c2 = anchor/230 - no n, mass ladder")
print("  T3 (Dark Matter):  f_c3 = anchor*(13/19)/230 - no n, mass ladder")
print("=" * 80)

```

How to Run

1. Save the code as `gut_complete_v2.py`
2. Install dependencies:

```
bash
pip install numpy matplotlib pandas
```

3. Run the program:

```
bash
python gut_complete_v2.py
```

Summary of Changes from the Original

Original Code	Updated Code
T2 uses <code>n_target = -exponent_n</code>	T2 uses <code>f_c2 = anchor/230</code> and mass ladder resonance
T3 uses <code>n_target_T3 = exponent_n × (13/19)</code>	T3 uses <code>f_c3 = anchor × (13/19) / 230</code> and mass ladder resonance
T2/T3 handshake used <code>cutoff_hz</code> generic	T2 handshake uses <code>anchor/230</code> ; T3 uses <code>anchor × (13/19) / 230</code>
Coherence_score assumed all branches are n-based	Coherence_score treats T2/T3 as frequency ladder-based